COMPUTER NETWORKS

UNT – III

NETWORK LAYER

The network layer or layer 3 of the OSI (Open Systems Interconnection) model is concerned delivery of data packets from the source to the destination across multiple hops or links. It is the lowest layer that is concerned with end − to − end transmission. The designers who are concerned with designing this layer needs to cater to certain issues. These issues encompass the services provided to the upper layers as well as internal design of the layer.

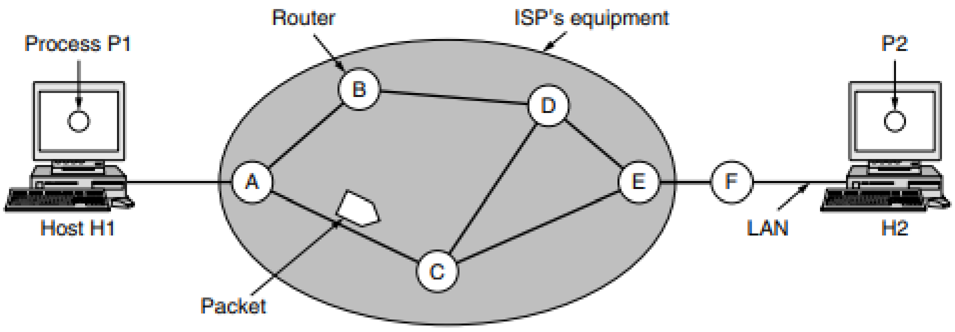


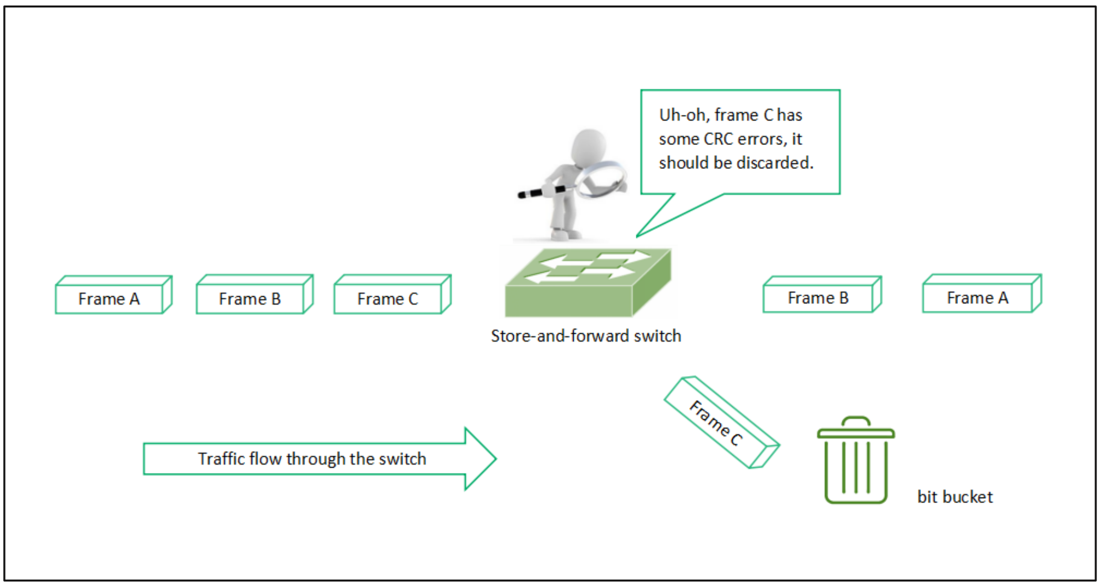
Figure 5-1. The environment of the network layer protocols.

The design issues can be elaborated under four heads −

- Store − and − Forward Packet Switching
- Services to Transport Layer
- Providing Connection Oriented Service
- Providing Connectionless Service

Store − and − Forward Packet Switching

The network layer operates in an environment that uses store and forward packet switching. The node which has a packet to send, delivers it to the nearest router. The packet is stored in the router until it has fully arrived and its checksum is verified for error detection. Once, this is done, the packet is forwarded to the next router. Since, each router needs to store the entire packet before it can forward it to the next hop, the mechanism is called store − and − forward switching.



Services to Transport Layer

The network layer provides service its immediate upper layer, namely transport layer, through the network − transport layer interface. The two types of services provided are −

- Connection − Oriented Service − In this service, a path is setup between the source and the destination, and all the data packets belonging to a message are routed along this path.
- Connectionless Service − In this service, each packet of the message is considered as an independent entity and is individually routed from the source to the destination.

The objectives of the network layer while providing these services are −

- The services should not be dependent upon the router technology.
- The router configuration details should not be of a concern to the transport layer.
- A uniform addressing plan should be made available to the transport layer, whether the network is a LAN, MAN or WAN.

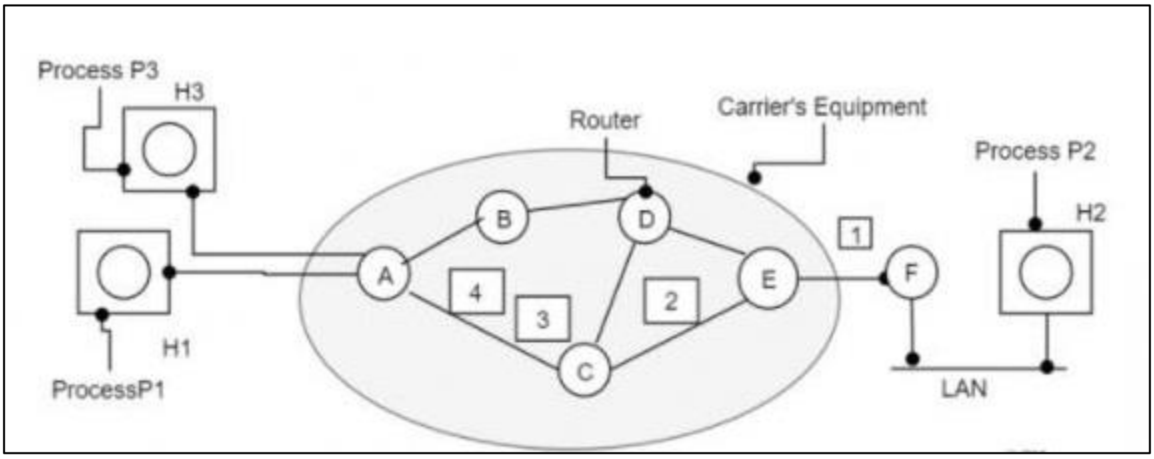
Providing Connection Oriented Service

We need a virtual-circuit subnet for connection-oriented service. Virtual circuits were designed to avoid having to choose a new route for every packet sent.

Instead, a route from the source machine to the destination machine is chosen as part of the connection setup and stored in tables inside the routers, when a connection is established. That route is utilised for all traffic flowing over the connection, and exactly the same manner even telephone works.

The virtual circuit is also terminated, when the connection is released. In connection-oriented service, every packet will have an identifier which tells to which virtual circuit it belongs to.

The implementation of connection-oriented services is diagrammatically represented as follows −



Example

Consider the scenario as mentioned in the above figure.

Step 1 − Host H1 has established connection 1 with host H2, which is remembered as the first entry in every routing table.

Step 2 − The first line of A's infers when packet is having connection identifier 1 is coming from host H1 and has to be sent to router W and given connection identifier as 1.

Step 3 − Similarly, the first entry at W routes the packet to Y, also with connection identifier 1.

Step 4 − If H3 also wants to establish a connection to H2 then it chooses connection identifier 1 and tells the subnet to establish the virtual circuit. This will be appearing in the second row in the table.

Step 5 − Note that we have a conflict here because although we can easily distinguish connection 1 packets from H1 from connection 1 packet from H3, W cannot do this.

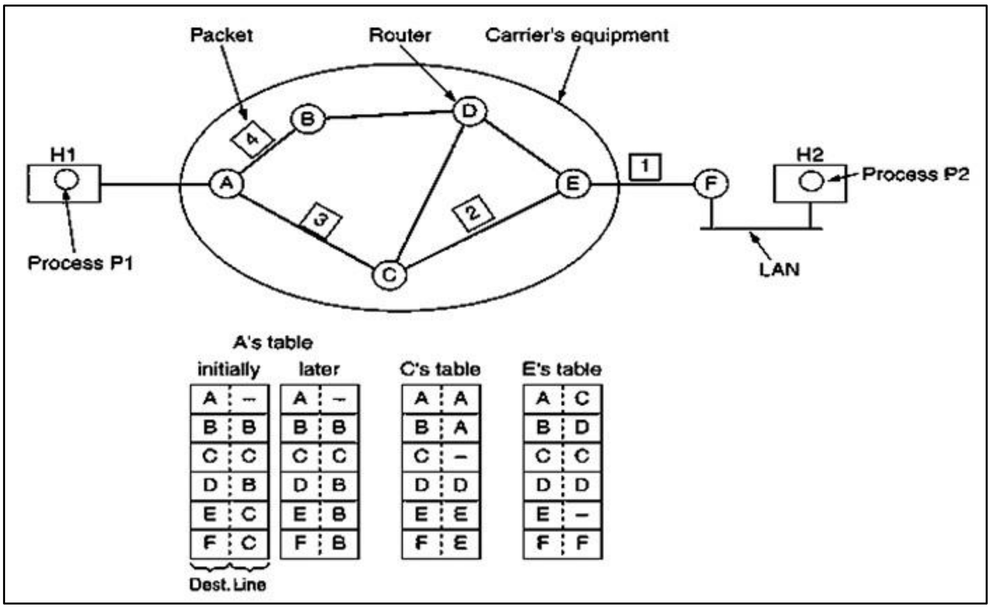
Step 6 − For this reason, we assign a different connection identifier to the outgoing traffic for the second connection. Avoiding conflicts of this kind is why routers need the ability to replace connection identifiers in outgoing packets. In some contexts, this is called label switching.

Providing Connectionless Service

When connectionless service is offered, packets are frequently called Datagrams (just like telegrams) because individual packets are injected to the subnet and are routed individually.

No advance setup is required. Subnets are called Datagram subnets. When Connection oriented service is provided, then before any packet is sent a path from source router to destination router is established. This connection is called Virtual Circuit and the subnet is called Virtual Circuit subnet.

The implementation of connectionless service is diagrammatically represented as follows −



A's table initially		A's table later		C's table		E's table	
A	–	A	–	A	A	A	C
B	B	B	B	B	A	B	D
C	C	C	C	C	–	C	C
D	B	D	B	D	D	D	D
E	C	E	B	E	E	E	–
F	C	F	B	F	E	F	F

Datagram Network

Let us discuss how datagram network works in stepwise manner −

Step 1 − Suppose there is a process P1 on host H1 and is having a message to deliver to P2 on host H2. P1 hands the message to the transport layer along with instructions to be delivered to P2 on H2.

Step 2 − Transport Layer code is running on H1 and within the operating system. It prepends a transport header to the message and the end result is given to the network layer.

Step 3 − Let us assume for this example a packet which is four times heavier than the maximum size of the packet, then the packet is broken to four different packets and each of the packet is sent to the router A using point to point protocol and from this point career takes over.

Step 4 − Each router will have an internal table saying where packets to be sent. Every table entry is a pair consisting of a destination and outgoing line to use for that destination. Only directly connected lines can be used.

Step 5 − For example A has only two outgoing lines to B and C, therefore every incoming packet must be sent to one of these routers, even if the ultimate destination will be some other router.

Step 6 − As the packets arrived at A, packet 1,2,3 and 4 were stored in brief. Then every packet is moved to C as per A's table. Packet 1 is forwarded to E and then moved to F. When packet 1 is moved to F, then it will be encapsulated in a data link layer and sent to H2 over to LAN. Packet 2 and 3 will also follow the same route.

Step 7 − When packet 4 reaches A, then it was sent to router B, even if the destination was F. For some purpose A decided to send packet 4 through a different route. It was because of the traffic jam in ACE path and the routing table was updated. Routing Algorithm decides routes, makes routing decisions and manages routing tables.

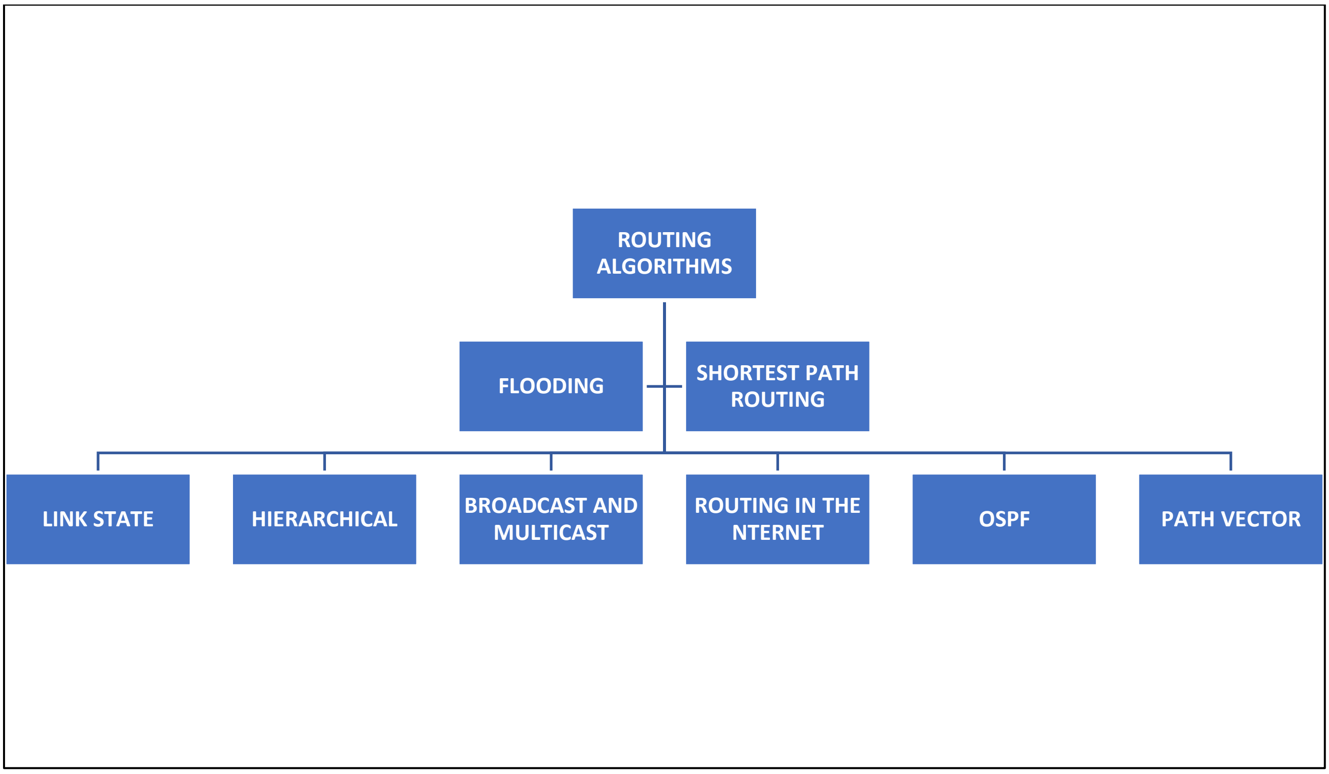
ROUTING ALGORITHM

Routing is the process of establishing the routes that data packets must follow to reach the destination. In this process, a routing table is created which contains information regarding routes that data packets follow. Various routing algorithms are used for the purpose of deciding which route an incoming data packet needs to be transmitted on to reach the destination efficiently.

Classification of Routing Algorithms

The routing algorithms can be classified as follows:

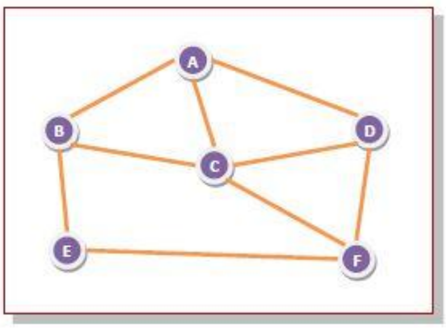
1. Adaptive Algorithms
2. Non-Adaptive Algorithms



1. FLOODING:

Flooding is a non-adaptive routing technique following this simple method: when a data packet arrives at a router, it is sent to all the outgoing links except the one it has arrived on.

For example, let us consider the network in the figure, having six routers that are connected through transmission lines.



Using flooding technique −

- An incoming packet to A, will be sent to B, C and D.
- B will send the packet to C and E.
- C will send the packet to B, D and F.
- D will send the packet to C and F.
- E will send the packet to F.
- F will send the packet to C and E.

Types of Flooding

Flooding may be of three types −

- **Uncontrolled flooding** − Here, each router unconditionally transmits the incoming data packets to all its neighbours.
- **Controlled flooding** − They use some methods to control the transmission of packets to the neighbouring nodes. The two popular algorithms for controlled flooding are Sequence Number Controlled Flooding (SNCF) and Reverse Path Forwarding (RPF).
- **Selective flooding** − Here, the routers don't transmit the incoming packets only along those paths which are heading towards approximately in the right direction, instead of every available paths.

Advantages of Flooding

- It is very simple to setup and implement, since a router may know only its neighbours.
- It is extremely robust. Even in case of malfunctioning of a large number routers, the packets find a way to reach the destination.
- All nodes which are directly or indirectly connected are visited. So, there are no chances for any node to be left out. This is a main criteria in case of broadcast messages.
- The shortest path is always chosen by flooding.

Limitations of Flooding

- Flooding tends to create an infinite number of duplicate data packets, unless some measures are adopted to damp packet generation.
- It is wasteful if a single destination needs the packet, since it delivers the data packet to all nodes irrespective of the destination.
- The network may be clogged with unwanted and duplicate data packets. This may hamper delivery of other data packets.

2. SHORTEST PATH ROUTING:

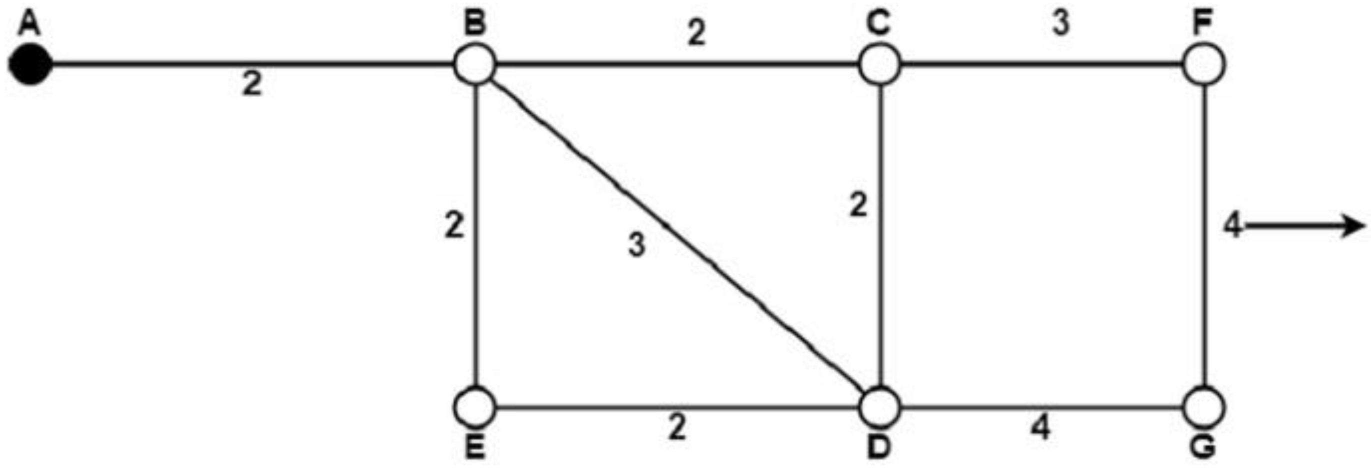
In this algorithm, to select a route, the algorithm discovers the shortest path between two nodes. It can use multiple hops, the geographical area in kilometres or labelling of arcs for measuring path length.

The labelling of arcs can be done with mean queuing, transmission delay for a standard test packet on an hourly basis, or computed as a function of bandwidth, average distance traffic, communication cost, mean queue length, measured delay or some other factors.

In shortest path routing, the topology communication network is defined using a directed weighted graph. The nodes in the graph define switching components and the directed arcs in the graph define communication connection between switching components. Each arc has a weight that defines the cost of sharing a packet between two nodes in a specific direction.

This cost is usually a positive value that can denote such factors as delay, throughput, error rate, financial costs, etc. A path between two nodes can go through various intermediary nodes and arcs. The goal of shortest path routing is to find a path between two nodes that has the lowest total cost, where the total cost of a path is the sum of arc costs in that path.

For example, Dijikstra uses the nodes labelling with its distance from the source node along the better-known route. Initially, all nodes are labelled with infinity, and as the algorithm proceeds, the label may change. The labelling graph is displayed in the figure.



Graphical Representation of Nodes with labeled path

It can be done in various passes as follows, with A as the source.

- Pass 1. B (2, A), C(∞,−), F(∞,−), e(∞,−), d(∞,−), G 60
- Pass 2. B (2, A), C(4, B), D(5, B), E(4, B), F(∞,−),G(∞,−)
- Pass 3. B(2, A), C(4, B), D(5, B), E(4, B), F(7, C), G(9, D)

We can see that there can be two paths between A and G. One follows through ABCFG and the other through ABDG. The first one has a path length of 11, while the second one has 9. Hence, the second one, as G (9, D), is selected. Similarly, Node D has also three paths from A as ABD, ABCD and ABED. The first one has a path length of 5 rest two have 6. So, the first one is selected.

All nodes are searched in various passes, and finally, the routes with the shortest path lengths are made permanent, and the nodes of the path are used as a working node for the next round.

3. LINK STATE ROUTING:

Link state routing is a technique in which each router shares the knowledge of its neighborhood with every other router in the internetwork.

The three keys to understand the Link State Routing algorithm:

- **Knowledge about the neighborhood:** Instead of sending its routing table, a router sends the information about its neighborhood only. A router broadcast its identities and cost of the directly attached links to other routers.
- **Flooding:** Each router sends the information to every other router on the internetwork except its neighbors. This process is known as Flooding. Every router that receives the packet sends the copies to all its neighbors. Finally, each and every router receives a copy of the same information.

- **Information sharing:** A router sends the information to every other router only when the change occurs in the information.

Link State Routing has two phases:

Reliable Flooding

- **Initial state:** Each node knows the cost of its neighbors.
- **Final state:** Each node knows the entire graph.

Route Calculation

Each node uses Dijkstra's algorithm on the graph to calculate the optimal routes to all nodes.

- The Link state routing algorithm is also known as Dijkstra's algorithm which is used to find the shortest path from one node to every other node in the network.
- The Dijkstra's algorithm is an iterative, and it has the property that after k^{th} iteration of the algorithm, the least cost paths are well known for k destination nodes.

Let's describe some notations:

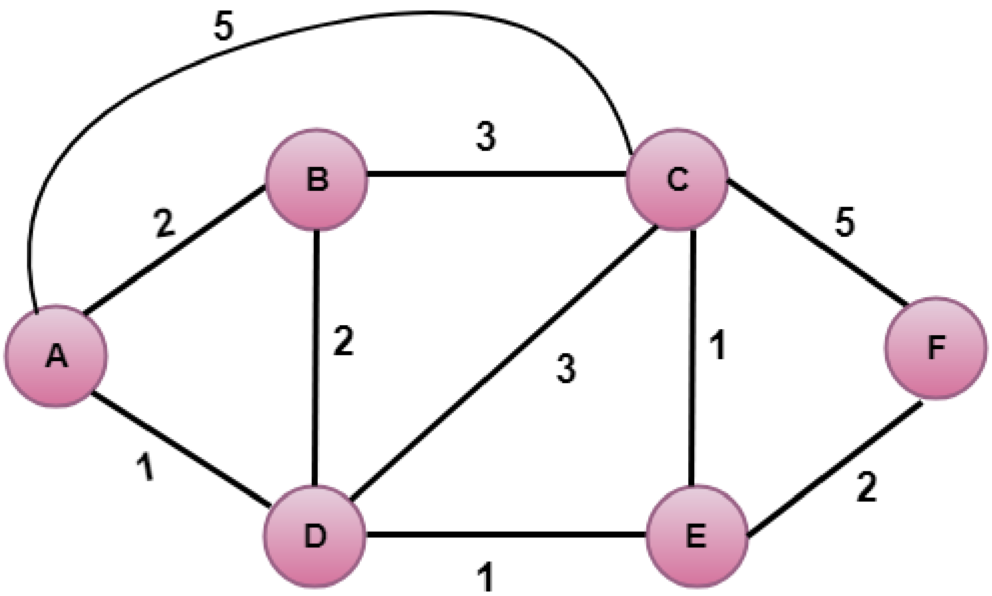
- **c(i , j):** Link cost from node i to node j. If i and j nodes are not directly linked, then c(i , j) = ∞.
- **D(v):** It defines the cost of the path from source code to destination v that has the least cost currently.
- **P(v):** It defines the previous node (neighbor of v) along with current least cost path from source to v.
- **N:** It is the total number of nodes available in the network.

Algorithm

```
Initialization
N = {A}     // A is a root node.
for all nodes v
if v adjacent to A
then D(v) = c(A,v)
else D(v) = infinity
loop
find w not in N such that D(w) is a minimum.
Add w to N
Update D(v) for all v adjacent to w and not in N:
D(v) = min(D(v) , D(w) + c(w,v))
Until all nodes in N
```

In the above algorithm, an initialization step is followed by the loop. The number of times the loop is executed is equal to the total number of nodes available in the network.

Let's understand through an example:



In the above figure, source vertex is A.

Step 1:

The first step is an initialization step. The currently known least cost path from A to its directly attached neighbors, B, C, D are 2,5,1 respectively. The cost from A to B is set to 2, from A to D is set to 1 and from A to C is set to 5. The cost from A to E and F are set to infinity as they are not directly linked to A.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞

Step 2:

In the above table, we observe that vertex D contains the least cost path in step 1. Therefore, it is added in N. Now, we need to determine a least-cost path through D vertex.

a) Calculating shortest path from A to B

1. v = B, w = D
2. D(B) = min(D(B) , D(D) + c(D,B))
3. = min(2, 1+2)>
4. = min(2, 3)
5. The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating shortest path from A to C

1. v = C, w = D
2. D(B) = min(D(C) , D(D) + c(D,C))
3. = min(5, 1+3)
4. = min(5, 4)
5. The minimum value is 4. Therefore, the currently shortest path from A to C is 4.</p>

c) Calculating shortest path from A to E

1. v = E, w = D
2. D(B) = min(D(E) , D(D) + c(D,E))
3. = min(∞, 1+1)
4. = min(∞, 2)
5. The minimum value is 2. Therefore, the currently shortest path from A to E is 2.

Note: The vertex D has no direct link to vertex E. Therefore, the value of D(F) is infinity.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞

Step 3:

In the above table, we observe that both E and B have the least cost path in step 2. Let's consider the E vertex. Now, we determine the least cost path of remaining vertices through E.

a) Calculating the shortest path from A to B.

1. v = B, w = E
2. D(B) = min(D(B) , D(E) + c(E,B))
3. = min(2 , 2+ ∞)
4. = min(2, ∞)
5. The minimum value is 2. Therefore, the currently shortest path from A to B is 2.

b) Calculating the shortest path from A to C.

1. v = C, w = E
2. D(B) = min(D(C) , D(E) + c(E,C))
3. = min(4 , 2+1)
4. = min(4,3)
5. The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

c) Calculating the shortest path from A to F.

1. v = F, w = E
2. D(B) = min(D(F) , D(E) + c(E,F))
3. = min(∞ , 2+2)
4. = min(∞ ,4)
5. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E

Step 4:

In the above table, we observe that B vertex has the least cost path in step 3. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through B.

a) Calculating the shortest path from A to C.

1. v = C, w = B
2. D(B) = min(D(C) , D(B) + c(B,C))
3. = min(3 , 2+3)
4. = min(3,5)
5. The minimum value is 3. Therefore, the currently shortest path from A to C is 3.

b) Calculating the shortest path from A to F.

1. v = F, w = B
2. D(B) = min(D(F) , D(B) + c(B,F))
3. = min(4, ∞)
4. = min(4, ∞)
5. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E

Step 5:

In the above table, we observe that C vertex has the least cost path in step 4. Therefore, it is added in N. Now, we determine the least cost path of remaining vertices through C.

a) Calculating the shortest path from A to F.

1. v = F, w = C
2. D(B) = min(D(F) , D(C) + c(C,F))
3. = min(4, 3+5)
4. = min(4,8)
5. The minimum value is 4. Therefore, the currently shortest path from A to F is 4.

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E
5	ADEBC					4,E

Final table:

Step	N	D(B),P(B)	D(C),P(C)	D(D),P(D)	D(E),P(E)	D(F),P(F)
1	A	2,A	5,A	1,A	∞	∞
2	AD	2,A	4,D		2,D	∞
3	ADE	2,A	3,E			4,E
4	ADEB		3,E			4,E
5	ADEBC					4,E
6	ADEBCF					

Disadvantage:

Heavy traffic is created in Line state routing due to Flooding. Flooding can cause an infinite looping, this problem can be solved by using Time-to-leave field

4. HIERARCHICAL ROUTING:

In hierarchical routing, the routers are divided into regions. Each router has complete details about how to route packets to destinations within its own region. But it does not have any idea about the internal structure of other regions.

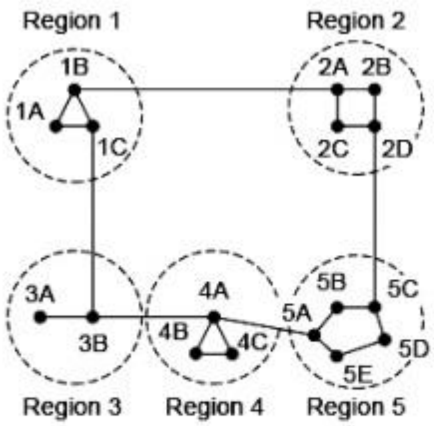
As we know, in both LS and DV algorithms, every router needs to save some information about other routers. When network size is growing, the number of routers in the network will increase. Therefore, the size of routing table increases, then routers cannot handle network traffic as efficiently. To overcome this problem we are using hierarchical routing.

In hierarchical routing, routers are classified in groups called regions. Each router has information about the routers in its own region and it has no information about routers in other regions. So, routers save one record in their table for every other region.

For huge networks, a two-level hierarchy may be insufficient hence, it may be necessary to group the regions into clusters, the clusters into zones, the zones into groups and so on.

Example

Consider an example of two-level hierarchy with five regions as shown in figure −



Let see the full routing table for router 1A which has 17 entries, as shown below −

Full Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2A	1B	2
2B	1B	3
2C	1B	3
2D	1B	4
3A	1C	3
3B	1C	2
4A	1C	3
4B	1C	4
4C	1C	4
5A	1C	4
5B	1C	5
5C	1B	5
5D	1C	6
5E	1C	5

When routing is done hierarchically then there will be only 7 entries as shown below −

Hierarchical Table for 1A

Dest.	Line	Hops
1A	-	-
1B	1B	1
1C	1C	1
2	1B	2
3	1C	2
4	1C	3
5	1C	4

Unfortunately, this reduction in table space comes with the increased path length.

Explanation

Step 1 − For example, the best path from 1A to 5C is via region 2, but hierarchical routing of all traffic to region 5 goes via region 3 as it is better for most of the other destinations of region 5.

Step 2 − Consider a subnet of 720 routers. If no hierarchy is used, each router will have 720 entries in its routing table.

Step 3 − Now if the subnet is partitioned into 24 regions of 30 routers each, then each router will require 30 local entries and 23 remote entries for a total of 53 entries.

Example

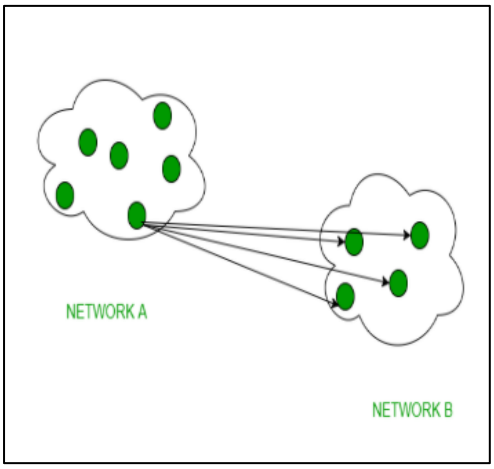
If the same subnet of 720 routers is partitioned into 8 clusters, each containing 9 regions and each region containing 10 routers. Then what will be the total number of table entries in each router.

Solution

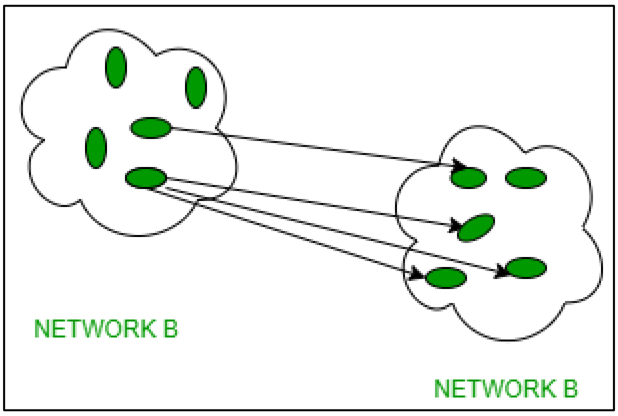
10 local entries + 8 remote regions + 7 clusters = 25 entries.

5. BROADCAST AND MULTICAST ROUTING:

1. Broadcast :
Broadcast transfer (one-to-all) techniques and can be classified into two types : Limited Broadcasting, Direct Broadcasting. In broadcasting mode, transmission happens from one host to all the other hosts connected on the LAN. The devices such as bridge uses this. The protocol like ARP implement this, in order to know MAC address for the corresponding IP address of the host machine. ARP does ip address to mac address translation. RARP does the reverse.



2. Multicasting :
Multicasting has one/more senders and one/more recipients participate in data transfer traffic. In multicasting traffic recline between the boundaries of unicast and broadcast. It server's direct single copies of data streams and that are then simulated and routed to hosts that request it. IP multicast requires support of some other protocols such as IGMP (Internet Group Management Protocol), Multicast routing for its working. And also in Classful IP addressing Class D is reserved for multicast groups.



Difference between Broadcast and Multicast :

S.No.	Broadcast	Multicast
1.	It has one sender and multiple receivers.	It has one or more senders and multiple receivers.
2.	It sent data from one device to all the other devices in a network.	It sent data from one device to multiple devices.
3.	It works on star and bus topology.	It works on star, mesh, tree and hybrid topology.

4.	It scale well across large networks.	It does not scale well across large networks.
5.	Its bandwidth is wasted.	It utilizes bandwidth efficiently.
6.	It has one-to-all mapping.	It has one-to-many mapping.
7.	Hub is an example of a broadcast device.	Switch is an example of a multicast device.

6. ROUTING IN THE INTERNET ROUTING:

Routing

- A Router is a process of selecting path along which the data can be transferred from source to the destination. Routing is performed by a special device known as a router.
- A Router works at the network layer in the OSI model and internet layer in TCP/IP model
- A router is a networking device that forwards the packet based on the information available in the packet header and forwarding table.
- The routing algorithms are used for routing the packets. The routing algorithm is nothing but a software responsible for deciding the optimal path through which packet can be transmitted.
- The routing protocols use the metric to determine the best path for the packet delivery. The metric is the standard of measurement such as hop count, bandwidth, delay, current load on the path, etc. used by the routing algorithm to determine the optimal path to the destination.
- The routing algorithm initializes and maintains the routing table for the process of path determination.

Routing Metrics and Costs

Routing metrics and costs are used for determining the best route to the destination. The factors used by the protocols to determine the shortest path, these factors are known as a metric.

Metrics are the network variables used to determine the best route to the destination. For some protocols use the static metrics means that their value cannot be changed and for some other routing protocols use the dynamic metrics means that their value can be assigned by the system administrator.

The most common metric values are given below:

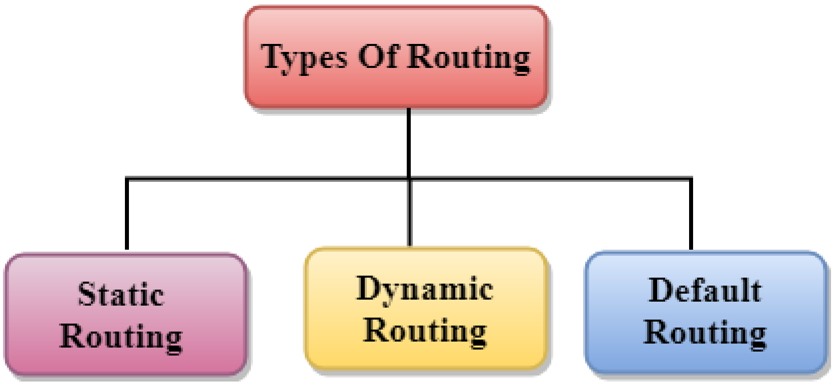
- **Hop count:** Hop count is defined as a metric that specifies the number of passes through internetworking devices such as a router, a packet must travel in a route to move from source to the destination. If the routing protocol considers the hop as a primary metric value, then the path with the least hop count will be considered as the best path to move from source to the destination.
- **Delay:** It is a time taken by the router to process, queue and transmit a datagram to an interface. The protocols use this metric to determine the delay values for all the links along the path end-to-end. The path having the lowest delay value will be considered as the best path.
- **Bandwidth:** The capacity of the link is known as a bandwidth of the link. The bandwidth is measured in terms of bits per second. The link that has a higher transfer rate like gigabit is preferred over the link that has the lower capacity like 56 kb. The protocol will determine the bandwidth capacity for all the links along the path, and the overall higher bandwidth will be considered as the best route.
- **Load:** Load refers to the degree to which the network resource such as a router or network link is busy. A Load can be calculated in a variety of ways such as CPU utilization, packets processed per second. If the traffic increases, then the load value will also be increased. The load value changes with respect to the change in the traffic.

- **Reliability:** Reliability is a metric factor may be composed of a fixed value. It depends on the network links, and its value is measured dynamically. Some networks go down more often than others. After network failure, some network links repaired more easily than other network links. Any reliability factor can be considered for the assignment of reliability ratings, which are generally numeric values assigned by the system administrator.

Types of Routing

Routing can be classified into three categories:

- Static Routing
- Default Routing
- Dynamic Routing



Static Routing

- Static Routing is also known as Nonadaptive Routing.
- It is a technique in which the administrator manually adds the routes in a routing table.
- A Router can send the packets for the destination along the route defined by the administrator.
- In this technique, routing decisions are not made based on the condition or topology of the networks

Advantages Of Static Routing

Following are the advantages of Static Routing:

- **No Overhead:** It has ho overhead on the CPU usage of the router. Therefore, the cheaper router can be used to obtain static routing.
- **Bandwidth:** It has not bandwidth usage between the routers.
- **Security:** It provides security as the system administrator is allowed only to have control over the routing to a particular network.

Disadvantages of Static Routing:

Following are the disadvantages of Static Routing:

- For a large network, it becomes a very difficult task to add each route manually to the routing table.
- The system administrator should have a good knowledge of a topology as he has to add each route manually.

Default Routing

- Default Routing is a technique in which a router is configured to send all the packets to the same hop device, and it doesn't matter whether it belongs to a particular network or not. A Packet is transmitted to the device for which it is configured in default routing.
- Default Routing is used when networks deal with the single exit point.
- It is also useful when the bulk of transmission networks have to transmit the data to the same hp device.
- When a specific route is mentioned in the routing table, the router will choose the specific route rather than the default route. The default route is chosen only when a specific route is not mentioned in the routing table.

Dynamic Routing

- It is also known as Adaptive Routing.
- It is a technique in which a router adds a new route in the routing table for each packet in response to the changes in the condition or topology of the network.
- Dynamic protocols are used to discover the new routes to reach the destination.
- In Dynamic Routing, RIP and OSPF are the protocols used to discover the new routes.
- If any route goes down, then the automatic adjustment will be made to reach the destination.

The Dynamic protocol should have the following features:

- All the routers must have the same dynamic routing protocol in order to exchange the routes.
- If the router discovers any change in the condition or topology, then router broadcast this information to all other routers.

Advantages of Dynamic Routing:

- It is easier to configure.
- It is more effective in selecting the best route in response to the changes in the condition or topology.

Disadvantages of Dynamic Routing:

- It is more expensive in terms of CPU and bandwidth usage.
- It is less secure as compared to default and static routing.

7. Path Vector Routing

Path Vector Routing is a routing algorithm in unicast routing protocol of network layer, and it is useful for interdomain routing. The principle of path vector routing is similar to that of distance vector routing. It assumes that there is one node in each autonomous system that acts on behalf of the entire autonomous system is called Speaker node .The speaker node in an AS creates a routing cable and advertises to the speaker nodein the neighbouring ASs .A speaker node advertises the path, not the metrics of the nodes, inits autonomous system or other autonomous systems

Dest	Path
A1	AS1
A2	AS1
A3	AS1
A4	AS1
A5	AS1

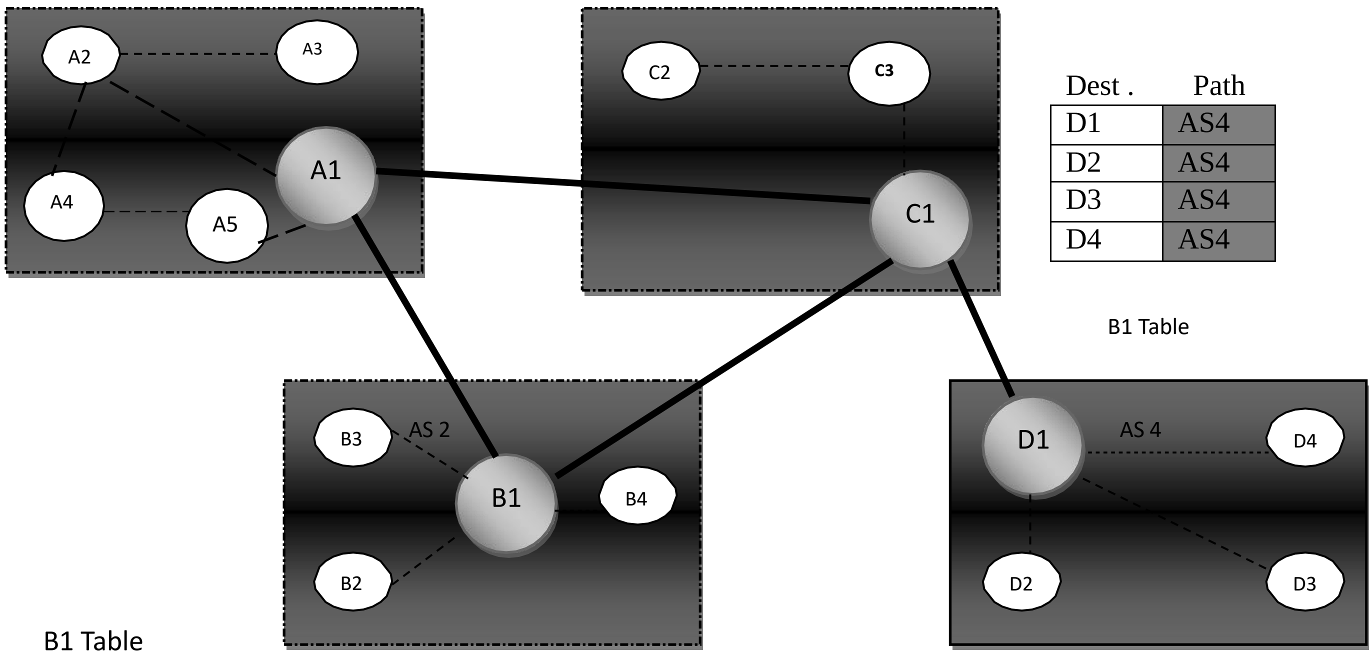
A1 Table

Dest .	Path
C1	AS3
C2	AS3
C3	AS3

C1 Table

AS 1

AS 3



Dest .	Path
D1	AS4
D2	AS4
D3	AS4
D4	AS4

B1 Table

B1 Table

Initial routing tables in path vector routine

It is the initial table for each speaker node in a system made four ASs. Here Node A1 is the speaker node for AS1, B1 for AS2, C1 for AS3 and D1 for AS4, Node A1 creates an initial table that shows A1 to A5 and these are located in AS1, it can be reached through it.

A speaker in an autonomous system shares its table with immediate neighbors ,here Node A1 share its table with nodes B1 and C1 , Node C1 share its table with nodes A1,B1 and D1 , Node B1 share its table with nodes A1 and C1 , Node D1share its table with node C1

If router A1 receives a packet for nodes A3 , it knows that the path is in AS1,but if it receives a packet for D1,it knows that the packet should go from AS1,to AS2 and then to AS3 ,then the routing table shows that path completely on the other hand if the node D1 in AS4 receives a packet for node A2,it knows it should go through AS4,AS3,and AS1,

FUNCTIONS

- PREVENTION OF LOOP

The creation of loop can be avoided in path vector routing .A router receives a message it checks to see if its autonomous system is in the path list to the destination if it is looping is involved and the message is ignored

- POLICY ROUTING

When a router receives a messages it can check the path, if one of the autonomous system listed in the path against its policy, it can ignore its path and destination it does not update its routing table with this path or it does not send the messages to its neighbours.

- OPTIMUM PATH

A path to a destination that is the best for the organization that runs the autonomous system

BORDER GATEWAY PROTCOL (BGP)

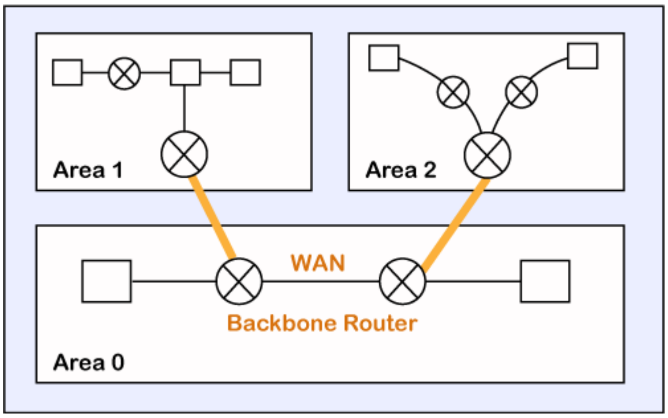
It first introduced in 1989, Is an interdomain routing protocol usingpath vector routing. It has three versions

1. STUB AS
2. MULTHIHOMED AS
3. TRANSIT AS

8. OSPF:

The OSPF stands for **Open Shortest Path First**. It is a widely used and supported routing protocol. It is an intradomain protocol, which means that it is used within an area or a network. It is an interior gateway protocol that has been designed within a single autonomous system. It is based on a link-state routing algorithm in which each router contains the information of every domain, and based on this information, it determines the shortest path. The goal of routing is to learn routes. The OSPF achieves by learning about every router and subnet within the entire network. Every router contains the same information about the network. The way the router learns this information by sending LSA (Link State Advertisements). These LSAs contain information about every router, subnet, and other networking information. Once the LSAs have been flooded, the OSPF stores the information in a link-state database known as LSDB. The main goal is to have the same information about every router in an LSDBs.

OSPF Areas



OSPF divides the autonomous systems into areas where the area is a collection of networks, hosts, and routers. Like internet service providers divide the internet into a different autonomous system for easy management and OSPF further divides the autonomous systems into Areas.

Routers that exist inside the area flood the area with routing information

In Area, the special router also exists. The special routers are those that are present at the border of an area, and these special routers are known as Area Border Routers. This router summarizes the information about an area and shares the information with other areas.

How does OSPF work?

There are three steps that can explain the working of OSPF:

Step 1: The first step is to become OSPF neighbors. The two connecting routers running OSPF on the same link creates a neighbor relationship.

Step 2: The second step is to exchange database information. After becoming the neighbors, the two routers exchange the LSDB information with each other.

Step 3: The third step is to choose the best route. Once the LSDB information has been exchanged with each other, the router chooses the best route to be added to a routing table based on the calculation of SPF.

THE NETWORK LAYER IN THE INTERNET

At the network layer, the Internet can be viewed as a collection of subnetworks or autonomous systems that are connected together. There is no real structure, but several major backbones exist. These are constructed from high-bandwidth lines and fast routers. Attached to the backbones are regional (mid-level) networks and attached to these regional networks are the LANs at many universities, companies, and Internet service providers.

1. IP PROTOCOL

The Internet Protocol (IP) is a protocol, or set of rules, for routing and addressing packets of data so that they can travel across networks and arrive at the correct destination. Data traversing the Internet is divided into smaller pieces, called packets. IP information is attached to each packet, and this information helps routers to send packets to the right place. Every device or domain that connects to the Internet is assigned an IP address, and as packets are directed to the IP address attached to them, data arrives where it is needed.

Once the packets arrive at their destination, they are handled differently depending on which transport protocol is used in combination with IP. The most common transport protocols are TCP and UDP.

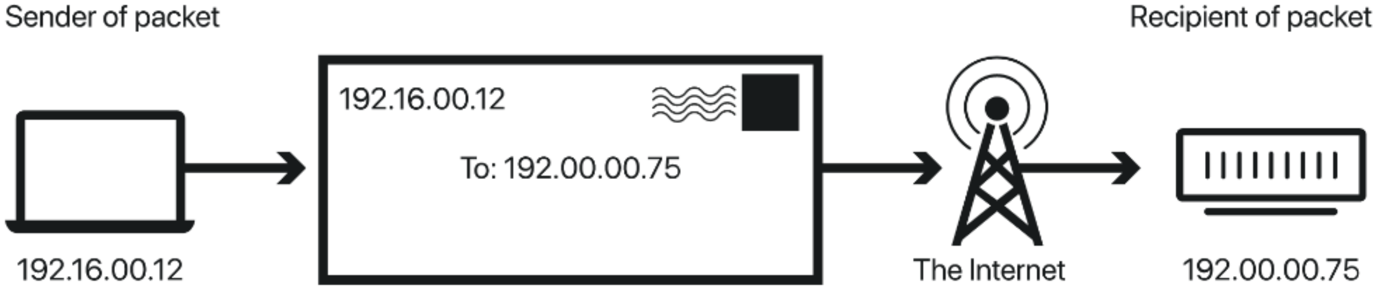
What is a network protocol?

In networking, a protocol is a standardized way of doing certain actions and formatting data so that two or more devices are able to communicate with and understand each other.

To understand why protocols are necessary, consider the process of mailing a letter. On the envelope, addresses are written in the following order: name, street address, city, state, and zip code. If an envelope is dropped into a mailbox with the zip code written first, followed by the street address, followed by the state, and so on, the post office won't deliver it. There is an agreed-upon protocol for writing addresses in order for the postal system to work. In the same way, all IP data packets must present certain information in a certain order, and all IP addresses follow a standardized format.

What is an IP address? How does IP addressing work?

An IP address is a unique identifier assigned to a device or domain that connects to the Internet. Each IP address is a series of characters, such as '192.168.1.1'. Via DNS resolvers, which translate human-readable domain names into IP addresses, users are able to access websites without memorizing this complex series of characters. Each IP packet will contain both the IP address of the device or domain sending the packet and the IP address of the intended recipient, much like how both the destination address and the return address are included on a piece of mail.



IPv4 vs. IPv6

The fourth version of IP (IPv4 for short) was introduced in 1983. However, just as there are only so many possible permutations for automobile license plate numbers and they have to be reformatted periodically, the supply of available IPv4 addresses has become depleted. IPv6 addresses have many more characters and thus more permutations; however, IPv6 is not yet completely adopted, and most domains and devices still have IPv4 addresses.

What is an IP packet?

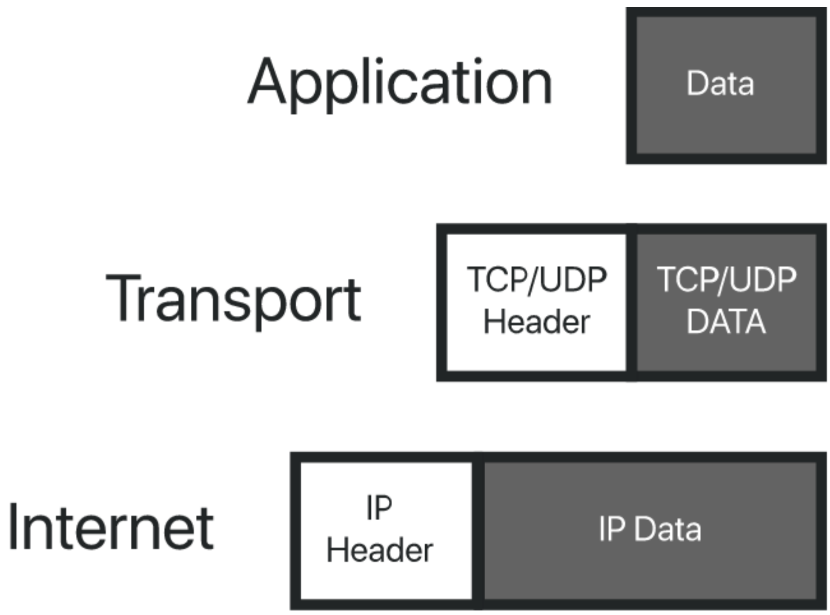
IP packets are created by adding an IP header to each packet of data before it is sent on its way. An IP header is just a series of bits (ones and zeros), and it records several pieces of information about the packet, including the sending and receiving IP address. IP headers also report:

- Header length
- Packet length
- Time to live (TTL), or the number of network hops a packet can make before it is discarded
- Which transport protocol is being used (TCP, UDP, etc.)

In total there are 14 fields for information in IPv4 headers, although one of them is optional.

How does IP routing work?

The Internet is made up of interconnected large networks that are each responsible for certain blocks of IP addresses; these large networks are known as autonomous systems (AS). A variety of routing protocols, including BGP, help route packets across ASes based on their destination IP addresses. Routers have routing tables that indicate which ASes the packets should travel through in order to reach the desired destination as quickly as possible. Packets travel from AS to AS until they reach one that claims responsibility for the targeted IP address. That AS then internally routes the packets to the destination.



Packets can take different routes to the same place if necessary, just as a group of people driving to an agreed-upon destination can take different roads to get there.

What is TCP/IP?

The Transmission Control Protocol (TCP) is a transport protocol, meaning it dictates the way data is sent and received. A TCP header is included in the data portion of each packet that uses TCP/IP. Before transmitting data, TCP opens a connection with the recipient. TCP ensures that all packets arrive in order once transmission begins. Via TCP, the recipient will acknowledge receiving each packet that arrives. Missing packets will be sent again if receipt is not acknowledged.

TCP is designed for reliability, not speed. Because TCP has to make sure all packets arrive in order, loading data via TCP/IP can take longer if some packets are missing.

TCP and IP were originally designed to be used together, and these are often referred to as the TCP/IP suite. However, other transport protocols can be used with IP.

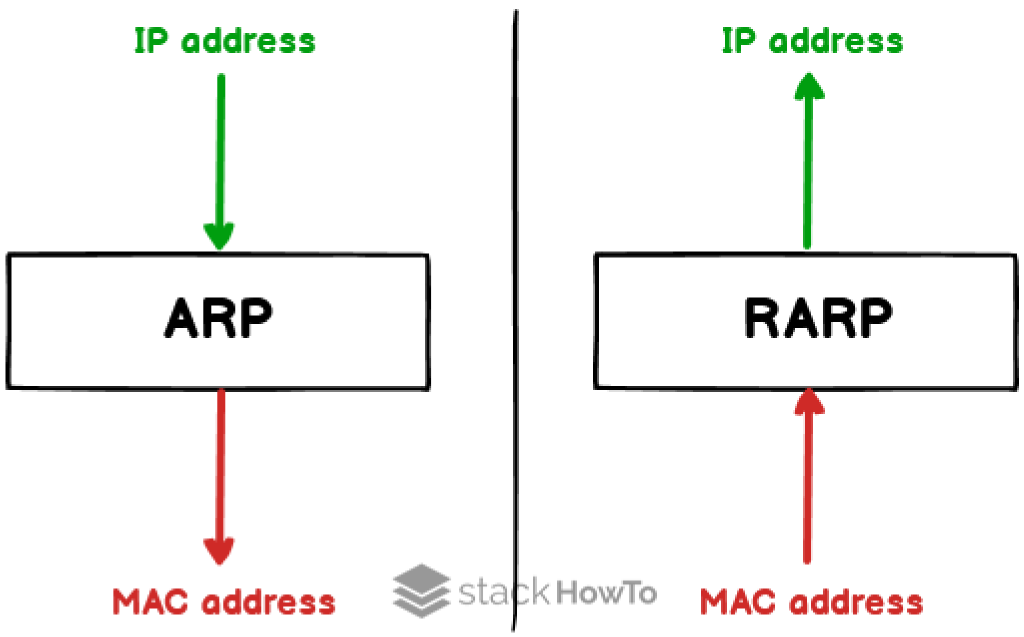
What is UDP/IP?

The User Datagram Protocol, or UDP, is another widely used transport protocol. It is faster than TCP, but it is also less reliable. UDP does not make sure all packets are delivered and in order, and it does not establish a connection before beginning or receiving transmissions.

2. ARP / RARP

ARP and ***RARP*** protocols are the main forms of ***LAN (Local Area Network)*** protocols. When a transmitter transfers an IP datagram from one host to another host, it necessities both the logical and physical addresses of the receiver. The dynamic mapping protocol supports two protocols: ARP protocol and RARP protocol. The primary distinction between these ARP and RARP protocols is that the ARP protocol acquires the receiver's physical address when given the receiver's logical address. In contrast, the RARP protocol obtains the host's logical address from the server when given the physical address of the host.

In this article, you will learn about the difference between ***ARP*** and ***RARP*** Protocols. But before discussing the differences, you must know about ARP and RARP Protocols with their advantages and disadvantages.



What is ARP?

ARP is commonly ***"Address Resolution Protocol"***. It is a form of ***network layer protocol***. As the ARP protocol is a dynamic mapping protocol, every host on the network is aware of another logical address of the host server. If the host requires transferring an ***IP datagram*** to another host, the IP datagram must be enclosed into a frame. It must be accomplished in such a manner that the datagram may easily flow across the physical network that exists between the receiver and the transmitter.

What is RARP?

RARP is commonly known as a ***"Reverse Address Resolution Protocol"***. The ***RARP protocol*** is also a type of ***network layer protocol***. A TCP/IP protocol enables any host to get its true IP address from the network server. As the name defines, the RARP protocol is an extended version of the ARP. It is usually the opposite of the ARP protocol.

Let us see that the difference between ARP and RARP that are as follows:

ARP	RARP
A protocol used to map an IP address to a physical (MAC) address	A protocol used to map a physical (MAC) address to an IP address
To obtain the MAC address of a network device when only its IP address is known	To obtain the IP address of a network device when only its MAC address is known
Client broadcasts its IP address and requests a MAC address, and the server responds with the corresponding MAC address	Client broadcasts its MAC address and requests an IP address, and the server responds with the corresponding IP address
IP addresses	MAC addresses
Widely used in modern networks to resolve IP addresses to MAC addresses	Rarely used in modern networks as most devices have a pre-assigned IP address
ARP stands for Address Resolution Protocol.	Whereas RARP stands for Reverse Address Resolution Protocol.

3. BOOTP:

Bootstrap Protocol (BOOTP) is a networking protocol which is used by networking administration to give IP addresses to each member of that network for participating with other networking devices by the main server.

Important Features of Bootstrap Protocol :
Here, we will discuss the features of Bootstrap Protocol as follows.

- Bootstrap Protocol (BOOTP) is a basic protocol that automatically provides each participant in a network connection with a unique IP address for identification and authentication as soon as it connects to the network. This helps the server to speed up data transfers and connection requests.
- BOOTP uses a unique IP address algorithm to provide each system on the network with a completely different IP address in a fraction of a second.
- This shortens the connection time between the server and the client. It starts the process of downloading and updating the source code even with very little information.
- BOOTP uses a combination of **TFTP** (Trivial File Transfer Protocol) and **UDP** (User Datagram Protocol) to request and receive requests from various network-connected participants and to handle their responses.

- In a BOOTP connection, the server and client just need an IP address and a gateway address to establish a successful connection. Typically, in a BOOTP network, the server and client share the same LAN, and the routers used in the network must support BOOTP bridging.
- A great example of a network with a TCP / IP configuration is the Bootstrap Protocol network. Whenever a computer on the network asks for a specific request to the server, BOOTP uses its unique IP address to quickly resolve them.

Uses of Bootstrap Protocol :
Here, we will discuss the uses of Bootstrap Protocol as follows.

1. Bootstrap (BOOTP) is primarily required to check the system on a network the first time you start your computer. Records the BIOS cycle of each computer on the network to allow the computer's motherboard and network manager to efficiently organize the data transfer on the computer as soon as it boots up.
2. BOOTP is mainly used in a diskless environment and requires no media as all data is stored in the network cloud for efficient use.
3. BOOTP is the transfer of a data between a client and a server to send and receive requests and corresponding responses by the networking server.
4. BOOTP supports the use of motherboards and network managers, so no external storage outside of the cloud network is required.

4. ICMP:

Internet Control Message Protocol (ICMP) is a network layer protocol used to diagnose communication errors by performing an error control mechanism. Since IP does not have an inbuilt mechanism for sending error and control messages. It depends on Internet Control Message Protocol(ICMP) to provide error control.

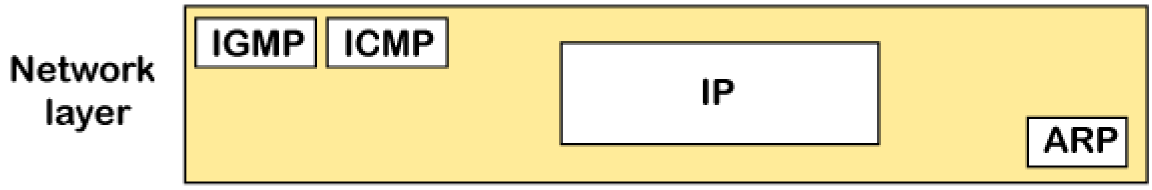
ICMP is used for reporting errors and management queries. It is a supporting protocol and is used by network devices like routers for sending error messages and operations information. For example, the requested service is not available or a host or router could not be reached.

Uses of ICMP

ICMP is used for error reporting if two devices connect over the internet and some error occurs, So, the router sends an ICMP error message to the source informing about the error. For Example, whenever a device sends any message which is large enough for the receiver, in that case, the receiver will drop the message and reply back ICMP message to the source.

Another important use of ICMP protocol is used to perform network diagnosis by making use of traceroute and ping utility. We will discuss them one by one.

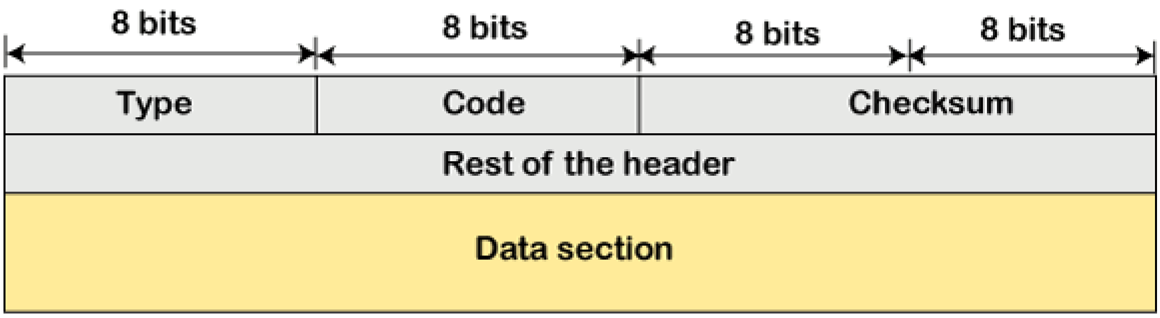
The ICMP resides in the IP layer, as shown in the below diagram.



ICMP Message Format

The message format has two things; one is a category that tells us which type of message it is. If the message is of error type, the error message contains the type and the code. The type defines the type of message while the code defines the subtype of the message.

The ICMP message contains the following fields:



- **Type:** It is an 8-bit field. It defines the ICMP message type. The values range from 0 to 127 are defined for ICMPv6, and the values from 128 to 255 are the informational messages.
- **Code:** It is an 8-bit field that defines the subtype of the ICMP message
- **Checksum:** It is a 16-bit field to detect whether the error exists in the message or not.

5. <u>DHCP:</u>

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to nay device, or node, on a network so they can communicate using IP (Internet Protocol). DHCP automates and centrally manages these configurations. There is no need to manually assign IP addresses to new devices. Therefore, there is no requirement for any user configuration to connect to a DHCP based network.

DHCP can be implemented on local networks as well as large enterprise networks. DHCP is the default protocol used by the most routers and networking equipment. DHCP is also called RFC (Request for comments) 2131.

DHCP does the following:

- DHCP manages the provision of all the nodes or devices added or dropped from the network.
- DHCP maintains the unique IP address of the host using a DHCP server.
- It sends a request to the DHCP server whenever a client/node/device, which is configured to work with DHCP, connects to a network. The server acknowledges by providing an IP address to the client/node/device.

DHCP is also used to configure the proper subnet mask, default gateway and DNS server information on the node or device.

There are many versions of DCHP are available for use in IPV4 (Internet Protocol Version 4) and IPV6 (Internet Protocol Version 6).

How DHCP works

DHCP runs at the application layer of the TCP/IP protocol stack to dynamically assign IP addresses to DHCP clients/nodes and to allocate TCP/IP configuration information to the DHCP clients. Information includes subnet mask information, default gateway, IP addresses and domain name system addresses.

DHCP is based on client-server protocol in which servers manage a pool of unique IP addresses, as well as information about client configuration parameters, and assign addresses out of those address pools.

The DHCP lease process works as follows:

- First of all, a client (network device) must be connected to the internet.
- DHCP clients request an IP address. Typically, client broadcasts a query for this information.
- DHCP server responds to the client request by providing IP server address and other configuration information. This configuration information also includes time period, called a lease, for which the allocation is valid.
- When refreshing an assignment, a DHCP clients request the same parameters, but the DHCP server may assign a new IP address. This is based on the policies set by the administrator.

Components of DHCP

When working with DHCP, it is important to understand all of the components. Following are the list of components:

- **DHCP Server:** DHCP server is a networked device running the DCHP service that holds IP addresses and related configuration information. This is typically a server or a router but could be anything that acts as a host, such as an SD-WAN appliance.
- **DHCP client:** DHCP client is the endpoint that receives configuration information from a DHCP server. This can be any device like computer, laptop, IoT endpoint or anything else that requires connectivity to the network. Most of the devices are configured to receive DHCP information by default.
- **IP address pool:** IP address pool is the range of addresses that are available to DHCP clients. IP addresses are typically handed out sequentially from lowest to the highest.
- **Subnet:** Subnet is the partitioned segments of the IP networks. Subnet is used to keep networks manageable.
- **Lease:** Lease is the length of time for which a DHCP client holds the IP address information. When a lease expires, the client has to renew it.
- **DHCP relay:** A host or router that listens for client messages being broadcast on that network and then forwards them to a configured server. The server then sends responses back to the relay agent that passes them along to the client. DHCP relay can be used to centralize DHCP servers instead of having a server on each subnet.

Benefits of DHCP

There are following benefits of DHCP:

Centralized administration of IP configuration: DHCP IP configuration information can be stored in a single location and enables that administrator to centrally manage all IP address configuration information.

Dynamic host configuration: DHCP automates the host configuration process and eliminates the need to manually configure individual host. When TCP/IP (Transmission control protocol/Internet protocol) is first deployed or when IP infrastructure changes are required.

6. NAT:

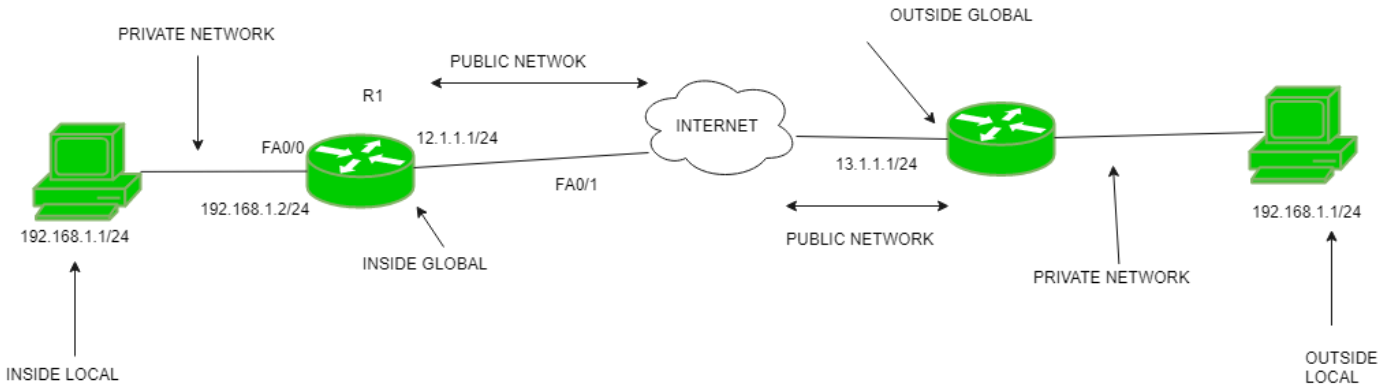
To access the Internet, one public IP address is needed, but we can use a private IP address in our private network. The idea of NAT is to allow multiple devices to access the Internet through a single public address. To achieve this, the translation of a private IP address to a public IP address is required. **Network Address Translation (NAT)** is a process in which one or more local IP address is translated into one or more Global IP address and vice versa in order to provide Internet access to the local hosts. Also, it does the translation of port numbers i.e. masks the port number of the host with another port number, in the packet that will be routed to the destination. It then makes the corresponding entries of IP address and port number in the NAT table. NAT generally operates on a router or firewall.

Network Address Translation (NAT) working –
Generally, the border router is configured for NAT i.e the router which has one interface in the local (inside) network and one interface in the global (outside) network. When a packet traverse outside the local (inside) network, then NAT converts that local (private) IP address to a global (public) IP address. When a packet enters the local network, the global (public) IP address is converted to a local (private) IP address.

If NAT runs out of addresses, i.e., no address is left in the pool configured then the packets will be dropped and an Internet Control Message Protocol (ICMP) host unreachable packet to the destination is sent.

Why mask port numbers ?
Suppose, in a network, two hosts A and B are connected. Now, both of them request for the same destination, on the same port number, say 1000, on the host side, at the same time. If NAT does only translation of IP addresses, then when their packets will arrive at the NAT, both of their IP addresses would be masked by the public IP address of the network and sent to the destination. Destination will send replies to the public IP address of the router. Thus, on receiving a reply, it will be unclear to NAT as to which reply belongs to which host (because source port numbers for both A and B are the same). Hence, to avoid such a problem, NAT masks the source port number as well and makes an entry in the NAT table.

NAT inside and outside addresses –
Inside refers to the addresses which must be translated. Outside refers to the addresses which are not in control of an organization. These are the network Addresses in which the translation of the addresses will be done.



- **Inside local address –** An IP address that is assigned to a host on the Inside (local) network. The address is probably not an IP address assigned by the service provider i.e., these are private IP addresses. This is the inside host seen from the inside network.

- **Inside global address –** IP address that represents one or more inside local IP addresses to the outside world. This is the inside host as seen from the outside network.

- **Outside local address –** This is the actual IP address of the destination host in the local network after translation.
- **Outside global address –** This is the outside host as seen from the outside network. It is the IP address of the outside destination host before translation.

Network Address Translation (NAT) Types –
There are 3 ways to configure NAT:

1. **Static NAT –** In this, a single unregistered (Private) IP address is mapped with a legally registered (Public) IP address i.e one-to-one mapping between local and global addresses. This is generally used for Web hosting. These are not used in organizations as there are many devices that will need Internet access and to provide Internet access, a public IP address is needed.

   Suppose, if there are 3000 devices that need access to the Internet, the organization has to buy 3000 public addresses that will be very costly.

2. **Dynamic NAT –** In this type of NAT, an unregistered IP address is translated into a registered (Public) IP address from a pool of public IP addresses. If the IP address of the pool is not free, then the packet will be dropped as only a fixed number of private IP addresses can be translated to public addresses.

   Suppose, if there is a pool of 2 public IP addresses then only 2 private IP addresses can be translated at a given time. If 3rd private IP address wants to access the Internet then the packet will be dropped therefore many private IP addresses are mapped to a pool of public IP addresses. NAT is used when the number of users who want to access the Internet is fixed. This is also very costly as the organization has to buy many global IP addresses to make a pool.

3. **Port Address Translation (PAT) –** This is also known as NAT overload. In this, many local (private) IP addresses can be translated to a single registered IP address. Port numbers are used to distinguish the traffic i.e., which traffic belongs to which IP address. This is most frequently used as it is cost-effective as thousands of users can be connected to the Internet by using only one real global (public) IP address.